

The Iterated Prisoner's Dilemma

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1 Introduction

The Prisoner's Dilemma is one of the most well-known problems in game theory. It describes a situation in which players must independently choose between cooperation and defection. Although cooperation can lead to a better outcome for both players, each player has an individual incentive to defect. This creates a conflict between individual incentives and the outcome that is collectively beneficial.

The Prisoner's Dilemma has been studied in many different forms, particularly when the game is played repeatedly. This paper will examine how different conditions of the game affect the outcomes observed in existing research. In particular, it will consider how factors such as the number of rounds, payoff structure, number of players, and strategies being used can influence the behaviour and performance observed in the game.

2 Background

2.1 The Prisoner's Dilemma

The Prisoner's Dilemma is a game in which players independently choose between two possible actions: cooperation and defection. The payoff received by each player depends on both their own action and the action chosen by the other player.

For a two-player Prisoner's Dilemma, the payoffs can be represented by the following matrix:

	C	D
C	(R, R)	(S, T)
D	(T, S)	(P, P)

where C represents cooperation and D represents defection. The four payoff values are usually required to satisfy

$$T > R > P > S.$$

Here, T is the temptation payoff received by a player who defects while the other player cooperates, R is the reward received when both players cooperate, P is the punishment received when both players defect, and S is the sucker's payoff received by a player who cooperates while the other player defects.

One commonly used set of payoffs is

$$T = 5, \quad R = 3, \quad P = 1, \quad S = 0.$$

Under these conditions, defection is a dominant strategy. Regardless of the action chosen by the other player, a player receives a higher payoff by defecting.

Therefore, when the game is played only once, both players choosing to defect forms a Nash equilibrium. However, mutual cooperation would give both players a higher payoff than mutual defection. This conflict between individual incentives and the collectively better outcome is the central dilemma of the game.

2.2 The Iterated Prisoner's Dilemma

The Prisoner's Dilemma can also be played repeatedly between the same players. This is known as the iterated Prisoner's Dilemma.

Instead of making a single decision, players make decisions over a sequence of rounds. The outcome of one round can therefore provide information that can be used when making a decision in a later round. This introduces the possibility that a player's current behaviour can depend on the previous behaviour of their opponent.

For example, a player may cooperate in an initial round and then continue cooperating if their opponent also cooperates. Alternatively, they may respond to a previous defection by defecting in the next round.

The repeated nature of the game therefore introduces strategies that are not relevant in the one-shot version. A strategy in the iterated Prisoner's Dilemma can be viewed as a rule that determines a player's action based on the information available from previous rounds.

2.3 Strategies in the Iterated Prisoner's Dilemma

Different strategies can produce very different behaviour when repeatedly interacting with one another.

Some simple examples are:

- **Always Cooperate:** the player cooperates in every round.
- **Always Defect:** the player defects in every round.
- **Tit-for-Tat:** the player cooperates in the first round and then copies the opponent's action from the previous round.

These strategies illustrate why the iterated version of the Prisoner's Dilemma is different from the one-shot version. For example, two players using Always Defect will defect in every round, while two players using Tit-for-Tat will cooperate in every round if they both begin by cooperating.

When different strategies interact, however, the sequence of outcomes can become more complicated. A strategy that performs well against one opponent may not perform equally well against another.

2.4 Why the Iterated Prisoner's Dilemma Is Interesting

The iterated Prisoner's Dilemma provides a way to study how behaviour changes when players interact repeatedly rather than only once.

One important area of interest is cooperation. Although defection is the dominant strategy in the one-shot game, repeated interaction can allow cooperation to persist through strategies that respond to previous behaviour.

Another important aspect is the interaction between strategies. The outcome of a game does not depend only on the strategy used by one player. It can also depend on the strategy used by the opponent. Consequently, the performance of a particular strategy may change when it is placed in a different strategic environment.

These observations raise questions about whether conclusions about the Prisoner's Dilemma are consistent across different ways of setting up the game.

2.5 Conditions Studied in Previous Research

Researchers have studied the Prisoner's Dilemma under a wide range of conditions. Rather than treating the game as having one fixed setup, different studies have changed various aspects of the game to investigate how these changes affect its outcomes.

Some of the conditions that have been varied include:

- **Number of rounds:** games may be played for a fixed number of rounds or under a probabilistic ending condition.
- **Payoff structure:** the values of T , R , P , and S can be changed, altering the incentives associated with cooperation and defection.
- **Number of players:** although the traditional Prisoner's Dilemma is a two-player game, researchers have also considered versions involving more players.
- **Strategies:** studies can compare different sets of strategies, which can affect the relative performance of the strategies being investigated.
- **Noise and errors:** some studies allow players to make errors, meaning that the action actually played may differ from the action intended by the player.
- **Ending conditions:** repeated games can differ in whether the number of rounds is known in advance or whether there is uncertainty about when the game will end.

The following sections examine research involving these different conditions and compare the effects that they have on the outcomes of the Prisoner's Dilemma.

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The number of rounds in an iterated Prisoner’s Dilemma determines how many opportunities players have to interact with one another. As the number of interactions increases, players have more opportunities to respond to their opponent’s previous actions, which can affect the development of cooperation and defection over time.

Mao et al. (2017) investigated the behaviour of players in repeated Prisoner’s Dilemma games over a much longer period than is typical in laboratory experiments. Their experiment involved 94 participants who played up to 400 repeated games, with each game consisting of 10 rounds. The experiment was conducted over 20 consecutive weekdays, allowing the researchers to observe how behaviour changed as participants gained experience with repeated interactions. In total, the participants made more than 374,000 individual decisions.

The researchers measured the cooperation rate at different rounds of the game and examined how these rates changed over the course of the experiment. They observed an “unravelling” effect during the first several days: the round at which players first defected tended to move earlier in the game. In particular, cooperation in the later rounds decreased initially. However, this decline did not continue indefinitely. After approximately one week, the unravelling process stabilized, and cooperation levels in the later rounds remained relatively stable for the remainder of the experiment.

This result demonstrates that the effect of repeated interaction cannot necessarily be understood by considering only the outcome of a single game. The cooperation rate can change as players accumulate experience across repeated games. Mao et al. also found that approximately 40% of participants behaved as “resilient cooperators”, continuing to cooperate even when doing so could result in a lower individual payoff. The authors then used a learning model to investigate how the presence of these resilient cooperators could affect the behaviour of the rest of the population.

The study therefore provides evidence that the behaviour observed in a repeated Prisoner’s Dilemma can depend on the amount of repeated interaction that players have experienced. It also shows that this relationship is not necessarily a simple increase or decrease in cooperation: cooperation initially became less stable as players gained experience, but eventually reached a more stable level. This suggests that the number and structure of repeated interactions can be important when interpreting experimental results from the Prisoner’s Dilemma.